Exam Date & Time: 11-Jan-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY 2024
SUBJECT: BIO 1071 / BIO_1071 - BIOLOGY FOR ENGINEERS

BIOLOGY FOR ENGINEERS [BIO 1071]

Marks: 50 **Duration: 180 mins.**

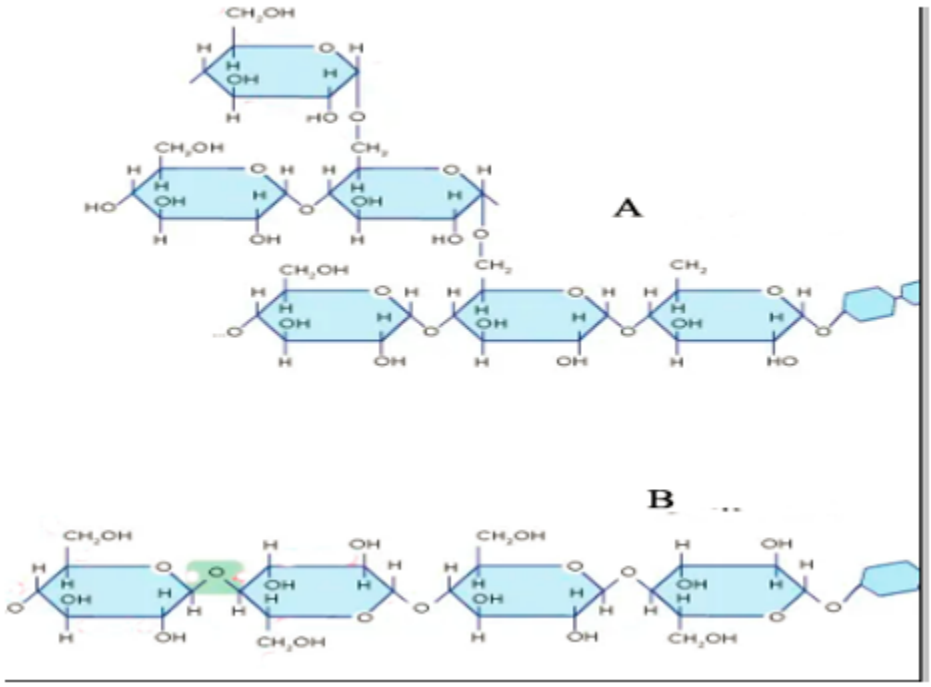
A

Answer all the questions.

1A) Living forms are build from polymers. What are the three important polymers of life and analyze their functions (3)

1B) According to you, what are the logical considerations that are significant towards elements of life? Justify (3)

1C) Analyse the following structures



(4)

i) What is the "engineering" in these structures making it different?
ii) According to you which is more ideal for fibril like arrangement? Justify
iii) Can the same enzyme break the bonds in A and B? Justify
(2+1+1 = 4 marks)

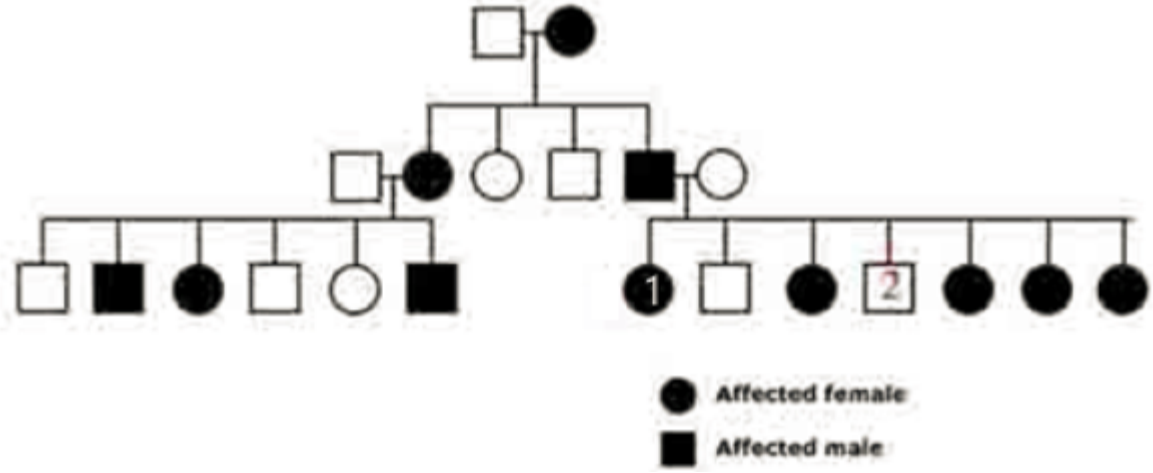
2A) Explain Mendel's monohybrid and dihybrid crosses logically (3)

2B) In Border Collies, black coat (B) is dominant to red coat (b). A breeder has a black male (3)

that has won numerous awards. The breeder would like to use the dog for breeding if he is pure breed. To learn this information, she testcrosses him with a red female (bb).

i) How the results will appear in the case of (a) Pure breed (b) Not pure breed? Justify
ii) If any of the puppies have red coat, what was the father's genotype?
(2+1 = 3 marks)

2C) Analyze the following pedigree



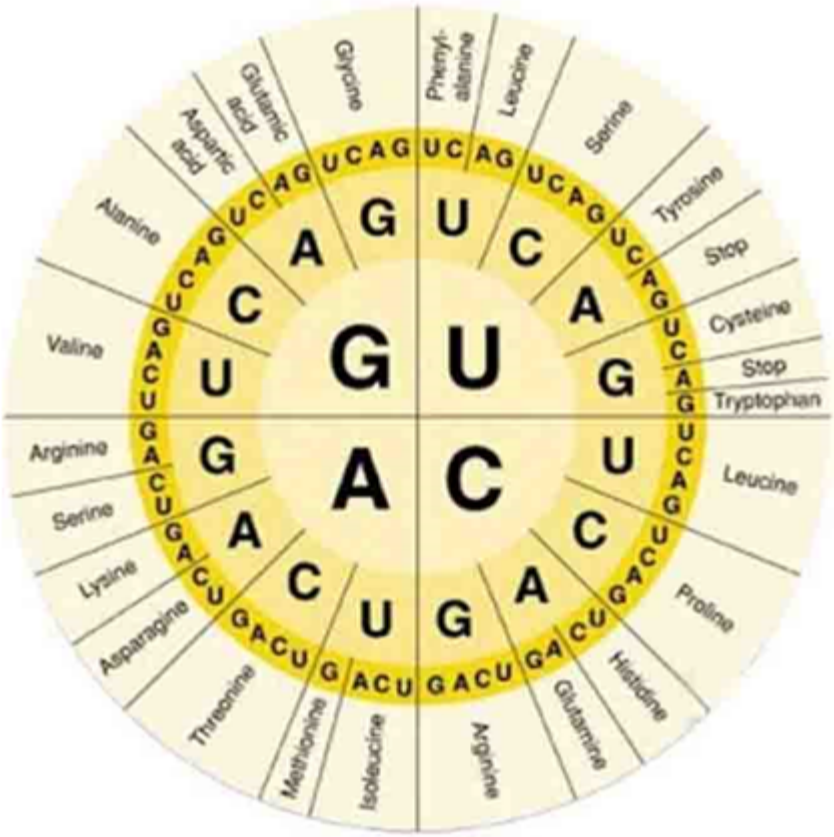
(4)

i) Give one logical reason for the above pedigree follows an X linked dominant pattern of inheritance
ii) X linked recessive mode is not possible in this case. Give a valid reason
iii) What are the genotypes of individual 1 and 2
iv) If the individual 2 marries an unaffected woman, what is the probability of their third child being affected?
(1+1+1+1 = 4 marks)

3A) Both DNA and RNA are coding for information, but how do they differ? (3)

3B) A non template DNA code 5'GTA 3'. Find the corresponding RNA code, anticodon code and the amino acid. Label the ends. Provide your answer in the following column format (3)

RNA code with labelled ends	
tRNA anticodon with labelled ends	
Corresponding amino acid	



3C) What can be the different modes of copying of DNA? (3 marks) How can it error free? (1 mark) (4)

4A) Both Darwinian and Lamarckian modalities of evolution appear to be important and reflect different aspects of the interaction between populations and the environment. Why then Lamarck's concept of evolution was limited in scope. (3)

4B) Mutation acts as a tool to change/improve the existing living systems. Logically explain (3 marks) (3)

4C) Consider hemoglobin, myoglobin, hemocyanin, leghaemoglobin, and chlorophyll. All of them are modified porphyrins. All these are modified from porphyrins. Explain logically 4 marks (4)

5A) All vaccines are not 100% effective, but some are. Justify logically (3 marks) (3)

5B) With a huge potential for innovation, biomimetics could evolve into a key process in businesses. The problem-driven process leads to Solution based in biomimetics. Give examples with explanation for the inspirations given below;
i) Self-Healing materials
ii) Inflatable structures
iii) Touch sensitive mechanism
(1+1+1 = 3 marks) (3)

5C) What is the concept for bioinspiration in the following cases? Explain. (4)

Bioinspiration	Concept	Explanation

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